# PROPOSAL FOR AIR TRAVEL OFFSET PROGRAM

Prepared for University of California
Office of the President

May 2015

Ron Hightower
Susan McMullan
Danielle Salah
Veena Patel

PRESIDIO
GRADUATE SCHOOL

# TABLE OF CONTENTS


EXECUTIVE SUMMARY ..... 1
- Current Business Process and Project Objectives ..... 1
- Sustainability Impacts ..... 1
- Methods ..... 1
- Recommendations ..... 1

INTRODUCTION ..... 2
- University Of California ..... 2
- Carbon Neutrality by 2025 ..... 2

PROJECT SCOPE ..... 2

AIR TRAVEL EMISSIONS ..... 2
- Current Reported Amounts ..... 2
- Pilot Spotlight: UCLA Air Travel Emissions ..... 3
- Behavior Change to Reduce Emissions ..... 3

DATA SOURCES AND METHODOLOGY FOR EMISSIONS TRACKING AND AIR TRAVEL REIMBURSEMENTS ..... 4
- Connexxus ..... 4
- Express ..... 4
- Methodology Used to Calculate Emissions ..... 5

TYPES OF OFFSET FEES ..... 6
- Flat Fee Per Trip ..... 6
- Tiered Flat Fee Per Trip ..... 6
- Fee Based on Miles ..... 6
- Fee Based on Price ..... 7

DEMAND FORECAST ..... 7

AIR TRAVEL OFFSETS ..... 9
- Verified Offsets ..... 9
- Local Offsets ..... 9

FUND INVESTMENTS IN OFFSETS ..... 9
- Fund Management ..... 9
- Management Structure ..... 10
- Type of Investment ..... 10

Project Evaluation ..... 10
Pilot Spotlight: Structure and Investments at UCLA ..... 10
MARKETING PLAN ..... 10
PROCESS DESCRIPTION ..... 11
STAKEHOLDERS ..... 12
STEPS TO IMPLEMENTATION ..... 13
BARRIERS TO IMPLEMENTATION ..... 13
OTHER OFFSET PROGRAMS ..... 14
University of Arizona ..... 14
Cornell University ..... 15
Seattle University ..... 15
Utah State University ..... 15
CONCLUSION ..... 16
ROLES AND RESPONSIBILITIES ..... 16
APPENDIX ..... 18
REFERENCES ..... 20

# EXECUTIVE SUMMARY

## Current Business Process and Project Objectives

The project is a pilot for the University of California, Los Angeles to offset greenhouse gas emissions (GHGs) from its business-related air travel by charging an offset fee through its electronic reimbursement system, Express. We discuss a method for calculating GHGs and offer four options for assessing the offset fee. We provide processes for the collected fees to create a fund to be used for local offset projects or to purchase verified offsets. Following the pilot period of approximately one year, the project would be expanded to other University of California (UC) campuses.

## Sustainability Impacts

In 2013, the most recent reporting year, UCLA business-related air travel contributed to at least 21,323 metric tons of GHGs--about five percent of UCLA's total scope 1, 2, and 3 emissions. The UC system has committed to carbon neutrality in its operations by 2025. Providing a means to offset GHGs from air travel through processes for collecting a fee and using the resulting fund to pay for local offset projects or purchase verified offsets will mitigate UCLA's GHG emissions from air travel.

## Methods

The team forecasted the total expected number of trips, mileage, and GHG emissions for 2015 and 2016 and the price per carbon ton in 2016. We then estimated the size of the fund needed to offset all air travel. To assess an air travel offset fee, the following types of calculation methods were considered:

1. Flat fee per trip
2. Tiered flat fee for California, domestic (non-California) and international travel
3. Fee based on miles and
4. Fee based on price of ticket.

## Recommendations

The team recommends that the air travel offset fee be assessed using a tiered flat fee structure that imposes a different and separate fee for trips (1) within California, (2) outside California but in the United States, and (3) involving international travel. We recommend a rollout process and that the UCLA pilot last from nine months to one year to provide adequate time for evaluation of its operations and benefits.

# INTRODUCTION

## University Of California

The University of California (UC) is a highly regarded public university system serving over 200,000 students in undergraduate, graduate, and professional programs. It has campuses in 10 cities and employs nearly 20,000 faculty and 136,000 staff. The UC Office of the President (UCOP) is located in Oakland and serves as the UC system's headquarters.

## Carbon Neutrality by 2025

Various groups at UC are focused on sustainability with a goal of carbon neutrality in operations by 2025. UC declared its commitment to this goal in 2007 when the then-president of UC signed the American College and University Presidents' Climate Commitment (ACUPCC) on behalf of the chancellors of each of the ten campuses. Campuses have accomplished emissions reductions through renewable energy, building efficiency, and local sustainable transportation, yet in order to meet this ambitious target, the UC system must address emissions from air travel.

# PROJECT SCOPE

The broad goal of this project is to significantly reduce greenhouse gas (GHGs) emissions from university-related air travel, which account for at least 10 percent of total GHGs, and to thereby contribute to the goal of carbon neutrality. To accomplish this, GreenEdge Consulting has created a pilot project to be implemented at UCLA to serve as a model that can easily be implemented UC system wide. The project will evaluate four options for assessing GHG emissions offset fees to provide funds for on campus offset projects.[1]

# AIR TRAVEL EMISSIONS

## Current Reported Amounts

Faculty, staff, and students engage in business-related air travel when they fly to meetings, conferences, study abroad programs, athletics, and similar activities. The purposes served are to exchange ideas, learn, make and maintain contacts, and for professional advancement and career development. Much of the air travel is mission critical or advances the UC system's goals of global engagement and leadership,

[1] The project as initially presented to GreenEdge was to evaluate GHG emissions due to air travel by University of California, San Francisco, faculty, staff, and students, and propose a tracking tool and program for offsets (as described in the Experiential Learning Program document provided when the project was chosen). The scope thereafter changed to providing a proposal to the UC Office of the President, and ultimately to the pilot project for UCLA described herein.

including professors travelling to conduct or present research. This limits how much air travel can be reduced and, to achieve neutrality, the UC system must implement an offset program. All ten campuses report their scope 1, 2, and 3 emissions to the ACUPCC Reporting System. For 2012, the total GHGs from air travel for all campuses was nearly 160,000 metric tons, representing on average 10 percent of total GHGs.[2] The percentage of GHGs attributable to air travel ranged from 30 percent at UC Santa Barbara to 5 percent at UC Davis, Los Angeles, and Riverside. (See Appendix, University of California GHG Reports)

## Pilot Spotlight: UCLA Air Travel Emissions

The project defines "UCLA-related air travel" as any travel made by faculty or staff as representatives of UCLA. This includes, but is not limited to, serving on government panels, attending conferences, and providing expert testimony. Based on survey data collected by UCLA, in 2013, faculty and staff there had nearly 107 million UCLA-related air travel miles. (See Appendix, UCLA Air Travel Emission Statistics Comparison over the years)

## Behavior Change to Reduce Emissions

This plan does not include recommendations to reduce business-related air travel. We and our clients at UC recognize that this is an area of tension for faculty whose career advancement requires writing—and often presenting at conferences—papers and who benefit from developing personal contacts with other researchers in their fields. We believe, however, that eventually behavioral change that reduces the frequency and mileage of air travel will be a necessary and significant part of achieving UC's carbon neutrality goal. The chief drawback of the offset program is that it does not actually reduce emissions from air travel. As noted by Le Quéré and her colleagues at the Tyndall Centre for Climate Change Research, "continued use of offsetting instead of reductions in flights contributes to the continued growth in the aviation sector, which is inconsistent with mitigation targets post-2030."

In the longer term, we believe that business-related travel by UC faculty, staff, and students should be reduced. This will require a culture shift in how the university operates. Professors, other faculty, and some students benefit from traveling to conferences, where they present papers, make contacts, and exchange ideas and information with others in their fields. Currently this is the norm in academia. In the recent working paper by the Tyndall Centre for Climate Change Research, the authors propose that research-related travel be monitored, incentives be given to reduce air travel, and that there be "a fundamental change in the research culture to align the walk with the talk." (Le Quéré, et al., 2015)

Making this shift will require providing for continued opportunities to stimulate ideas and make personal contacts independent of flying to meetings, and the online alternatives

[2] 2012 was the most recent year in which the largest number of campuses submitted reports. The figures described come from 2012 reports for seven campuses, 2013 reports for two campuses, and a 2011 report for one campus (which was its most recent report).

must be instead of, rather than in addition to in-person conferences. (Le Quéré, et al., 2015) It will also be important to ensure that career development is not adversely affected. In a survey the Tyndall researchers conducted, a large percentage of respondents agreed that flying helped them to maintain working relationship and networks and that reducing flying would limit their career progression. (Ibid.)

## DATA SOURCES AND METHODOLOGY FOR EMISSIONS TRACKING AND AIR TRAVEL REIMBURSEMENTS

### Connexxus

Connexxus is a web-based travel booking site similar to Expedia or Priceline. The booking site was created exclusively for the UC system. Faculty, staff, and students are encouraged to use Connexxus to book hotels, car rentals and most importantly air travel. Increased participation rates correlates to higher discounts and group rates for travel accommodations. Though cost-effective incentives are provided, Connexxus is highly underutilized. Participation across all UC campuses varies, with an average utilization rate of 40% in 2012. UCLA's utilization rate rises just above the average coming in at about 50%.

The low participation rate can be attributed to consumers using alternative travel sites when booking travel arrangements. Expedia and Priceline are mainstream providers that are heavily advertised guaranteeing cheaper rates. In addition, these public sites are much easier to navigate and compare different travel arrangements and prices by vendor. We believe that increased awareness of Connexxus as the preferred vendor for travel would slightly increase participation. But to gain a significant increase in Connexxus participation, an incentive such a competition between departments with the winner receiving a monetary award or an expedited reimbursement turn around could be tested.

Despite some negative aspects of Connexxus, valuable data are calculated and stored. The key data extracted from Connexxus are miles traveled and CO2 emissions per flight. (It also tracks date booked and date traveled.) Because Connexxus is able to calculate CO2 emissions acceptable by UC standards, it is a great tool for tracking air travel emissions. However due to the low participation rate the majority of air travel emissions are not accounted for. Connexxus captures only an estimated 50% of UCLA-related air travel miles. We recommend increasing Connexxus participation to track and accurately gauge air travel emissions.

### Express

Express is an electronic reimbursement system used by UCOP, UC Merced, and UCLA. UCLA administers the Express system for all three locations. Express tracks information and reimburses hotels, meals, rental cars, supplies, equipment, and most importantly air travel. Approximately 90% of UCLA's air travel is reimbursed through Express; travel for athletics is the exception, as planes are chartered and reimbursed through accounts payable. Unfortunately, the data are not aggregated and it would require accessing

individual ticket information for each record to determine miles traveled. The scale at which this would need to be done makes it an impractical solution.

But since the majority of UCLA-related air travel passes through Express for reimbursement purposes, the system is ideal to assess an air travel emissions offset fee. Tracking the amount of air travel emissions, however, is not possible in Express, as key fields are missing such as miles traveled and CO2 emissions. The key information in Express is the reimbursement amount, date of reimbursement, and purpose of travel.

## Methodology Used to Calculate Emissions

The associated GHG emissions for air travel is calculated using the equation determined in a 2008 study by Kwan. A variety of factors were considered before the calculation method could be developed. Kwan noted the following assumptions:

Aircraft type was limited to three models which best represented the type of aircraft that would most likely be used, based on the type and distance of each trip. The models were: Boeing 737-700 for domestic flights within the United States, Boeing 777-300 for trans-Atlantic flights, and Boeing 747-400 for trans-Pacific flights.

When determining emissions due to the landing and takeoff cycles (LTOs)--any GHG producing activity that takes place below 1000 meters in altitude--he assumed only one LTO cycle per flight. This assumes all flights are direct and fails to account for any connections the traveler might experience.

Fuel economy values were calculated from data reported by the Bureau of Transportation Statistics for the three aircraft types, which provided a specific number within the range of values offered by Boeing for each aircraft model.

The GHG emission factor for aircrafts at cruising altitude was taken from the Intergovernmental Panel on Climate Change (IPCC) as constant across aircraft models.

Several assumptions needed to be made in order to calculate emissions per passenger. Because he calculated seat numbers based on aircraft specifications, he assumed all seats to be economy class. From there, he used average load (number of passengers per available seats) for domestic and international flights to calculate the number of passengers on a typical flight for these two distance categories.

Based on these assumptions, Kwan supplied an equation that can be used to calculate GHG emissions based on the number of miles traveled in each of the three flight categories: domestic, trans-Atlantic, and trans-Pacific. (Kwan, 2008)

Ultimately, this "Calvin Method," as we and our UC clients refer to it, was determined to be comparable to many widely used calculators. However, his "low and high carbon emission estimates" that were calculated using the lower and higher input variables within their ranges were determined to be relatively far apart. This illustrates that the best way to calculate a more accurate value for GHG emissions related to air travel is to use higher quality data. We acknowledge the difficulties of obtaining such granular data but believe that the Calvin method is nonetheless an acceptable methodology for calculation given the current state of data collection.

## TYPES OF OFFSET FEES

We examined four types of calculation methods to assess an air travel offset fee, based on the capability of Express and other air travel offset program structures. These consisted of:

1. Flat fee per trip
2. Tiered flat fee for California, domestic (non-California) and international travel
3. Fee based on miles and
4. Fee based on price of ticket.

### Flat Fee Per Trip

This option ranked highest in terms of ease and lowest cost. No changes would need to be made to the Express user interface as the fee would be assessed on the back end. These back-end modifications would need to calculate, assess, collect and distribute the fee. Based on prior Express modifications done by the UCLA IT department, a conservative, rough, estimate of $20,000 was provided. In addition, the cost of yearly maintenance and adjustments for this added feature would be about $5,000. Assessing a flat fee per trip would not take into account the mileage or emissions generated by each trip. For example, a domestic one-hour flight would be assessed the same fee as an international 12-hour flight.

### Tiered Flat Fee Per Trip

To more accurately assess a fee based on miles and emissions, a tiered flat fee structure was developed. A separate fee for either California, domestic outside California, or international would be assessed. Express already has the ability to designate domestic and international flights. The Express user marks this required field when entering a travel expense for reimbursement. To add California to the system would cost around $6,000. A total of $30,000 is expected to be needed to get this option off the ground and running.

### Fee Based on Miles

A fee based on miles traveled would offer a more precise measure to offset air travel emissions per flight. Calculating CO2 emissions would be based on mileage and take into account the radiative force per passenger mile. Unfortunately, this would not be cost effective or user friendly. Since Express is a reimbursement program, it is not set up to track and calculate emissions based on miles traveled. Modifications to add this capability would be costly. Also, the Express user would need to enter raw data into Express such as origin and destination flights, miles, layovers, stops, airline, etc. to ensure accurate calculations. Because this would be burdensome, it is unlikely to be accepted by users and we do not recommend it as a viable option.

## Fee Based on Price

Since a fee based on actual miles traveled is not feasible using Express, an alternative way to calculate the fee is to take a percentage of the air travel reimbursement. This option would result in different fees for similar flights. Therefore this option would be able to generate a large fund but would not accurately assess a comparable true offset fee.

# DEMAND FORECAST

UCLA has compiled air travel data over the years from Connexxus, Anthony Travel Sports Agency, and faculty survey results. With this data 2015 and 2016 figures were projected using linear regression for total flights, emissions and miles. Figures for international flights average around 13% of total flights. This percentage was used to project 2015 and 2016 international and domestic flights. (Table 1)

**Table 1: Data on emissions, miles, flights for 2010-2014 and forecast for 2015-2016**

| | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 |
|---|---|---|---|---|---|---|---|
| Emissions | 15760 | 15612 | 19292 | 21323 | 21325 | 22384 | 23564 |
| Miles | 78836728 | 78025272 | 96129576 | 106857001 | 106348589 | 116280316 | 123759110 |
| Total Flights | 46397 | 48638 | 59055 | 61683 | 61759 | 66865 | 70894 |
| Domestic Flights | 40365 | 42315 | 51452 | 53742 | 53808 | 58173 | 61678 |
| International Filghts | 6032 | 6323 | 7603 | 7941 | 7951 | 8692 | 9216 |

Linear regression was used due to the gradual increase in air travel from 2007 to 2014. If air travel continues to increase at a steady pace the following trend lines can be used to predict the future figures. The R^2 figures indicate a high accuracy rate around 90% if air travel continues to increase.

**Table 2: R^2 for air travel forecast**

| | Trend Line | R^2 |
|---|---|---|
| Flight | =4029 * Years - 8051570 | 0.92 |
| Emissions | =1179 * Year - 2354668 | 0.88 |
| Miles | =7478793 * Year - 14953489479 | 0.89 |

Forecasting the price per carbon ton in 2016 three methods were evaluated, naive, 3 period moving average and 3 period weighted moving average. (Table 2) The California Air Resources Board's Carbon price per 1 metric ton of CO2e was retrieved from California Carbon Allowance Futures daily market price. (Daily Dashboard, 2015) The daily prices for each year were averaged to project the 2016 price of $13.54. With only 5 years of data this figure will not be precise, however with the increase over the past three years the price will continue to rise with a 2015 year end price of $13.00 and increasing another dollar in 2016.

**Table 3: Forecasting average carbon price per ton in 2016 using three methods**

| | | | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | MSE |
|---|---|---|---|---|---|---|---|---|---|
| Average Carbon Price Per Ton | | | $17.85 | $15.13 | $14.36 | $12.02 | $12.78 | 13.54 | |
| Naïve | | | | | $12.41 | $13.59 | $9.68 | $13.54 | $7.94 |
| 3 Period Moving Average | | | | | | $15.78 | $13.84 | $13.05 | $15.25 |
| 3 Period Weighted Moving Average | | | | | | $15.29 | $13.34 | $12.87 | $11.00 |
| 0.2 | 0.3 | 0.5 | | | | | | | |

The size of the fund needed to offset all air travel was estimated. (Table 4) These amounts will fluctuate depending on the current market price of carbon offsets and the number and distance of flights. The baseline to offset all air travel emissions in 2016 is projected to be $320,000. With the cost of Express modifications included this number would increase by $30,000. As Express only provides international and domestic categories for reimbursements, a California fee could not be generated for the tiered flat fee option (which provides for three different fee amounts), without system modifications. It is apparent, however, that an international flight generates more carbon emissions, which would cost more to purchase verified offsets. Note that the fee per mile calculation has been excluded as Express cannot accommodate these modifications. Also the fee per mile would not accurately assess a fee, so no assessment was projected.

**Table 4: Projected costs to offset flights in 2016 and fees per flight to fund offsets**

| | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 |
|---|---|---|---|---|---|---|
| Total Emissions | 15612 | 19292 | 21323 | 21325 | 22384 | 23564 |
| Domestic Emissions | 9981 | 12273 | 13533 | 13479 | 14719 | 15659 |
| International Emisions | 5631 | 7019 | 7790 | 7846 | 7665 | 7905 |
| Total Offset Cost | $278,674 | $291,888 | $306,198 | $256,327 | $286,071 | $319,055 |
| Domestic Offset Cost | $178,154 | $185,683 | $194,334 | $162,015 | $188,113 | $212,017 |
| International Offset Cost | $100,520 | $106,204 | $111,865 | $94,312 | $97,958 | $107,039 |
| Flat Fee Per Flight | $5.73 | $4.94 | $4.96 | $4.15 | $4.28 | $4.50 |
| Flat Fee Per Domestic | $4.21 | $3.61 | $3.62 | $3.01 | $3.23 | $3.44 |
| Flat Fee Per International | $15.90 | $13.97 | $14.09 | $11.86 | $11.27 | $11.61 |

## AIR TRAVEL OFFSETS

To address the UC system's air travel emissions, efforts are needed to reduce the amount of business-related air travel, along with offsetting current air travel. An apt analogy is the dual need when improving sustainability of an energy portfolio to invest in both energy efficiency and renewable energy. This dual approach is necessary for the UC system. The remainder of this proposal focuses on creating an offset fund for verified and local offsets.

### Verified Offsets

Verified offsets can be purchased from businesses or non-profit organizations to balance GHG emissions released during air travel. Offsets most often represent an investment elsewhere in an emissions-reducing activity, such as replacement of fossil fuels with renewable energy, capturing methane gas, and reforestation of land. The quality of offsets is highly variable since calculations are often indirect and projects are often completed around the world. Certifying organizations that verify offsets have improved the overall quality found in the market today. One such organization, The Climate Action Reserve, has developed protocols for eligibility and accounting for sixteen offsetting actions, such as forest, landfill, and livestock projects. These protocols ensure that projects are completed, permanent, and verified. Offsets are issued in the unit Climate Reserve Tonne (CRTs) which represent one metric ton of carbon reduction or sequestration. CRTs are currently purchased by the UC system to offset other activities, such as on-site power generation.

### Local Offsets

Offset monies can also be spent on local projects that reduce GHGs. For example, a campus could invest its offset fees in greening, sustainable transportation, renewable energy, energy efficiency, and any other activity that captures or reduces emissions. The emissions reduction must be "in addition" to the reductions that would occur without the offset investment. UC stakeholders stated a preference for local offset projects over purchasing verified offsets, in order to keep funds and benefits within the UC system and maintain oversight.

## FUND INVESTMENTS IN OFFSETS

### Fund Management

Across UC, the existing structure that reviews sustainability decisions differs by campus. Therefore, each campus, upon proposal of an offset program, must individually determine the following: fund management structure, type of investments, and criteria for evaluating investments. Fund management proposals must be reviewed and approved by UCOP before any offset fees are collected. UCOP will conduct annual reviews to determine whether the proposals are being successfully executed and

identifying any necessary changes. The entire offset program faces reputational risk if offset fees are improperly invested.

### Management Structure

Proposed management structures are encouraged to include a committee of campus stakeholders, either existing or created for this purpose. The committee could be comprised of faculty and students with sustainability and facility management staff serving as committee advisors. Alternatively, the committee could be a group of sustainability and facilities staff. Each proposal for an offset program must include the committee governance structure.

### Type of Investment

Fund management proposals should specify whether investments are designated for local offset projects, verified offsets, or a combination of verified and local offsets. If they are purchasing offsets, they must propose where they will source the offsets from.

### Project Evaluation

Project evaluation criteria should be stated in the proposal, and should include projected GHGs captured or offset, cost of project, ease of implementation, lifetime of investment, visibility on campus, facilitation of sustainable behavior change, and alignment with Sustainability Plan objectives. All projects must be in addition to planned investments.

### Pilot Spotlight: Structure and Investments at UCLA

After the fee is assessed and collected on all UCLA-related travel by Corporate Finance, the funds would be housed and managed by the UCLA Sustainability Committee. The UCLA Sustainability Committee is comprised of staff members representing the Chancellor's Office, Capital Programs, Facilities Management, Transportation, Environment Health and Safety, Recreation, Housing, Student Affairs, and University Communications. The UCLA Air Travel Offset program would invest in a combination of verified offsets through the Climate Action Reserve and on-campus projects. UCLA currently purchases offsets for power they generate, therefore there are few barriers to purchasing offsets through this source. The campus has capacity for on-campus investments, especially for renewable energy projects.

University stakeholders, including those at UCLA, stated a preference for local offset projects over purchasing verified offsets, in order to keep funds and benefits within the UC system and to maintain oversight. Offset investments could be spent in on-campus projects for greening, sustainable transportation, renewable energy, energy efficiency, and any other activity that captures or reduces carbon. However, local offset projects cost more and are more difficult to account for than purchase verified offsets.

## MARKETING PLAN

Once the pilot program is in effect, participation would be mandatory for all UCLA-related air travel. We, however, recommend a rollout process to ensure that the offset

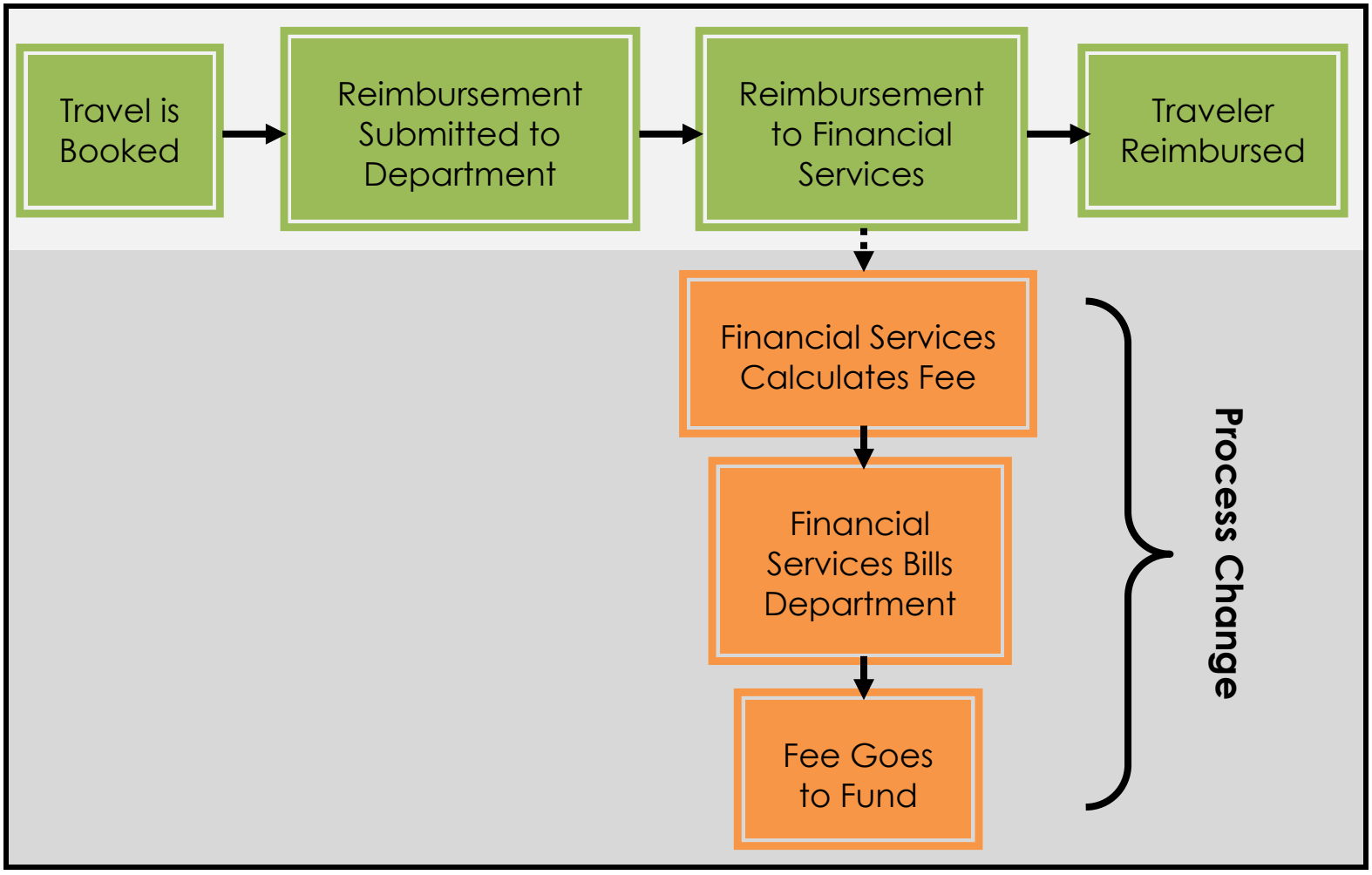
fee structure is amenable to faculty, staff, and students. Rollout efforts should begin during proposal development with stakeholder engagement. Possible sources for feedback include consultation with the Faculty Welfare Committee and student government representatives.

Before the offset fee is introduced, the UCLA Sustainability Committee (or the proposing party on campus) should distribute notice to faculty, staff, and students. In addition to the offset program type (local, verified, or combination) and fee structure, notice could include framing information on the call for carbon neutrality by 2025 and reducing GHG emissions from UCLA-related air travel.

In addition to promoting the offset program, resources to avoid air travel, such as videoconferencing systems, should also be highlighted to encourage behavior change, as discussed above, under **Behavior Change to Reduce Emissions**. To support the necessary culture shift in air travel and academia, campuses could create a webpage where staff who forgo air travel can submit their action and how it was accomplished (e.g., web conferencing).

After the program launches, UCLA should publicize which offsets were purchased and local projects funded from the offset fee. For on-campus projects, signage and other promotional materials are encouraged to increase awareness of GHG offsets.

## PROCESS DESCRIPTION



Travel Booking and Reimbursement

The current process is as follows:

1. Travel is booked (Connexxus or outside resource)
2. Individual engages in travel
3. Individual submits reimbursement form to department's Financial Services contact
4. Contact submits reimbursement form to Financial Services
5. Financial Services issues reimbursement to individual traveler

Fund Creation

Under GreenEdge's plan, the process would be revised at step four, wherein the following steps would be added:

6. Financial Services calculates associated fee
7. Financial Services bills department responsible for travel
8. Fees are collected into a UC system wide offset fund

## STAKEHOLDERS

UCOP has a significant role in sustainability efforts across campuses. Though many decisions can be made locally, some aspects of implementation require UCOP involvement and, to be most effective, certain initiatives should or must be centrally coordinated. The University of California Policy Advisory Committee "is charged with recommending a framework for system wide administrative policy development and maintenance, providing oversight and direction to executive policy owners, and assuring that policies are accessible, presented within a logical framework and appropriately disseminated." (Ethics, Compliance and Audit Services, n.d.) In making our recommendations, GreenEdge Consulting is aware of the role of the UC Policy Advisory Committee. We understand that this committee and the Travel Management Council have roles in approving implementation of an offset fee UC system wide.

Other UC system wide stakeholders with an interest in this project are the Sustainability Steering Committee, which is made up of administrative vice-chancellors for each campus; the Sustainable Transportation Working Group, which is under the steering committee; Climate Action Committee; and the Climate Change Working Group. At UCLA, the entities that are involved in this project— through providing data, other information and input, answering questions, and participating in weekly phone calls with GreenEdge Consulting—are Corporate Finance, Transportation, Sustainability, and subunits within those entities. All of them have a role in booking, processing and approving UCLA-related air travel miles, establishing an offset fee and decisions concerning use of funds generated by the fee. The Administrative Vice Chancellor for UCLA will determine the source and allocate monies needed to modify the reimbursement system and Financial Services will see that modifications are made to collect the fee.

## STEPS TO IMPLEMENTATION

A successful pilot project at UCLA is paramount to implementing an air travel offset fee UC system wide. Beginning at UCLA is a strategic decision because the Express system is as an easy entry point to collect the offset fee. Once the Express system has been updated to serve the needs of the UCLA pilot, it can be applied to UCOP and UC Merced, which also use Express. Before beginning the pilot at UCLA, there must be consultation with the affected stakeholders to review the necessary policy changes, as well as the proposed structure of the fund creation process. Work could begin concurrently on the necessary Express modifications so that the back end processes are in place while the program is under review.

Once launched, we recommend, based on input from our UC clients, that the pilot run for at least nine months (during the academic school year) and up to a full year. During the first year, the fund will be expected to cover the upfront costs of the required operational and technology changes. The total amount of fees collected should allow UCLA stakeholders to determine the effectiveness of the program, judged by its ability to offset air travel emissions through whichever means UCLA determines to use.

In addition to testing the viability of the fund on an operational level, the pilot will serve as a first chance to add GHG emissions from air travel to the discussion on sustainability and environmental stewardship that already exists on campus. We acknowledge that getting buy in for the program from all stakeholders, including participating faculty and staff, is crucial to wide range acceptance and success. The channels used to bring the message will be different at each campus, but we expect that UCLA will serve as an excellent example for sustainability messaging around this project. As part of this effort, we encourage discussion within a variety committees and meetings, especially those that exists system wide, such as the UC Sustainable Transportation Steering Committee and the UC Climate Action Committee.

## BARRIERS TO IMPLEMENTATION

For the pilot project, the funds for offsets will be collected from UCLA departments. One key challenge to implementation will be getting stakeholder buy-in to dedicate funds for offsets when department budgets may already be constrained. This barrier may be somewhat alleviated given that the funds are used for on-campus projects.

Travel to conduct research and present at conferences is encouraged for faculty to achieve tenure and raise the profile of academic departments. Therefore, implementing a fee for UCLA-related air travel miles creates a tension between this objective and UC system's directive to achieve carbon neutrality by 2050.

Another key barrier is the imprecision in quantifying local offset projects. The cost of a carbon capturing project can differ substantially based on project type, location, and responsible party. Furthermore, with local offset projects, it is difficult to account for the actual carbon sequestered. For example, the amount of carbon accounted for by an urban garden could change significantly season-to-season and year-to-year. Therefore, both aspects of the offset calculation--the cost and the amount of carbon sequestered--are variable figures. Purchasing verified offsets alleviates this challenge,

but UC stakeholders specify a preference for local offsets to keep the benefits of the funds collected within the UC system.

In contrast with purchasing verified offsets, creating a local offset program leaves the responsibility for project implementation with UCLA. Facilities Management would need to add the sequestering projects to their pipeline of campus improvements. If sequestering projects are not implemented in a timely manner or tracked and accounted, then the credibility of the offset program would be compromised.

Another area of disagreement that may impede implementation is whether offset projects meet the "additionality" requirement (to count as an offset the action must go beyond planned improvements). If projects are not considered offsets because they lack "additionality" they may be characterized as reduction efforts, but this may invalidate the grounds for creating and managing the fund.

## OTHER OFFSET PROGRAMS

We found little information about other universities' business-related air travel or their offset or reduction programs that is current, detailed, and comparable. Though many universities have a goal of reducing GHGs from air travel and other causes—which is not surprising given widespread participation in the ACUPCC—information is generally lacking about how much they offset or reduced from this specific source. Other universities report the same obstacles as UC: difficulty tracking distance with any precision, no information for some flights, and other lapses such that calculating actual GHGs is an approximation at best. Indeed, the University of Arizona (UA), after stating that air travel and resulting GHG emissions decreased approximately 28% from fiscal years 2011 to 2013, noted that air travel data confidence is low because averages, extrapolations, and assumptions are used to derive the data, and methodologies used in calculating the air miles changed over the two-year reporting period. (Greenhouse Gas Emissions Report Fiscal Year 2013, 2013) We discuss four universities below.

### University of Arizona

For rough comparison purposes, UA reported that emissions from air travel accounts for 15.43 percent of the UA carbon footprint—more than student, faculty and staff commuting combined. (Treading Lightly) UA's financial system tracking shows that faculty, staff and students travel approximately 5 million passenger miles. UA acknowledges, however, that not all travel is tracked. They are working to develop a voluntary carbon-offset purchasing system that is integrated into the process of purchasing individual airline tickets using the financial system. Similar to one part of the pilot we are proposing for UCLA, it would pool funds to purchase offset credits—possibly from local or regional projects in large quantities. UA states that pooling funds in this way would maximize purchasing power with a single request for proposals.

### Cornell University

In 2014, Cornell faculty and students were responsible for 30,000 tons of carbon from air travel. (Compare to UCLA's more than 21,000 tons reported for 2013.) This makes up 13% of their total GHGs. (Greenhouse Gas Emissions Inventory, n.d.) The university is engaged in efforts to reduce this amount by decreasing business travel when possible and purchasing carbon offsets. Cornell has partnered with a local carbon offset program that charges $3.00 per trip and uses the funds to invest in energy efficiency projects for low income households. One project offsets about 100 tons of CO2 per year. (Carbon Neutral Travel Policy, n.d.)

### Seattle University

Air travel makes up 38% of emissions generated by Seattle University (SU), and half of it is due to athletics. (Toward a More Sustainable and Just World, 2014)To reduce air travel emissions, the university's account payable group has begun to use a shared spreadsheet to more accurately track air travel miles and calculate the resulting emissions. Eventually SU plans to require all employees to book air travel through a specific vendor. Until it becomes mandatory, SU will target the booking arrangement to departments with high air travel. (PCS Meeting MInutes, 2014) In addition, SU is attempting to reduce air travel by promoting its web conferencing room. The university will evaluate purchasing carbon offsets in 2020. (The President's Committee on Sustainability, 2011)

### Utah State University

Utah State University (USU) reports that faculty, staff and students traveling on USU business logged over 17 million miles in air travel in 2012, which, with along with auto travel, accounts for 19 percent of its carbon emissions. To address this problem, USU developed an offset program that is processed through the travel reimbursement system, similar to the UCLA pilot program we developed. Its Sustainability Council added to their reimbursement forms, a means to make a voluntary $10 donation to fund on campus projects. The donation is deducted from reimbursements and allocated to a carbon offset fund. USU collected $9,000 in 2012, the first year of operation. The money was used to upgrade to an electronic fuel injection mower. The following year, funds were used for energy efficiency renovations and projects addressing air quality improvements.

Using its travel reimbursement form, financial systems, and database and information technology modifications, USU has developed a voluntary carbon-offset program in which travelers may check a box on their travel reimbursement form, indicating their intent to deduct $10 from their reimbursement as a tax-exempt donation to on-campus projects to reduce USU's carbon output. (Utah State University Carbon Offset Program, n.d.)

Though many colleges and universities acknowledge that air travel is a significant part of their overall GHG emissions, there is very little progress in identifying the need for and creating an offset program. The UC system will be among the first universities to establish an air travel GHG offset program.

## CONCLUSION

GreenEdge Consulting reviewed data from UCLA on its business-related air travel and forecast the number of flights, mileage, and GHGs for 2015 and 2016. We determined average carbon price per ton in 2016 using three methods and projected the total needed to offset flights in 2016 and amount of the fee per flight to fund those offsets. Four options for determining how to assess the fee were considered. GreenEdge Consulting recommends a tiered flat fee to correspond most closely with actual GHGs, although this will require an additional upfront costs to modify the reimbursement system and modest ongoing maintenance costs.

We determined the cost of modifications to UCLA's travel reimbursement to allow for a separate fee per flight to be charged for travel at three flight distances: California, domestic outside California, and international.

GreenEdge Consulting outlined the operational processes for such a reimbursement system and the creation and award of money from the offset fund. Across the UC system, campuses differ in their ability to invest offsets on campus, structures in place to manage program, and preference for type of offsets purchased. Therefore, each campus is to define these elements in their proposal to UCOP to begin an air travel offset program. Given the directive for carbon neutrality by 2025, we recommend that stakeholders engaged for this process continue to develop and launch the pilot program at UCLA in the next year. Formulating and testing the offset program at UCLA will remove barriers for other campuses to launch their program. However, campuses have significantly different reimbursement processes, therefore the process diagram and understanding billing software capabilities will need to be revisited at other campuses. Merced and UCOP also utilize Express, therefore may be fitting locations to expand the program.

Finally, though our recommendations do not include a reduction in business-related air travel, i.e., flying less, we believe that UC will eventually need to move in this direction. Program suggestions to communicate air travel emissions may aid reduction efforts, yet a cultural shift in the expectation of faculty and staff to travel for career advancement is ultimately needed.

## ROLES AND RESPONSIBILITIES

All team members of GreenEdge Consulting participated in one in-person and numerous telephone meetings with representatives from UCOP and UCLA (and on one occasion, UC's Sustainable Transportation Working Group) to identify the scope of this project and gain needed information about policies, processes, and data. **Ron Hightower** researched and wrote sections on offset options and other offset programs and performed data analysis for the demand forecast, price per carbon ton, and size of the fund needed to offset all air travel. **Susan McMullan** researched and wrote sections on project scope, stakeholders, behavior change, and other offset programs, wrote the executive summary and conclusion, and served as editor and team contact with the client.

**Veena Patel** researched and wrote sections on air travel offsets, fund investments, and marketing plan. Veena took the lead on the Powerpoint presentation and writing a proposal to present our research at the California Higher Education Sustainability Conference (CHESC). **Danielle Salah** researched and wrote sections on data sources, methods to calculate emissions, steps to implementation, and process description. **Team GreenEdge** will be presenting "Exploring Offset Options for UC Air Travel" at the CHESC Conference in July 2015.

## APPENDIX

| | University of California GHG Reports | | |
|---|---|---|---|
| | 2012 GHGs air travel | 2012 GHGs total | Percent from air travel |
| Berkeley | 21,100 | 138,791 | 15% |
| Davis | 13,415 | 268,168 | 5% |
| Irvine | 29,680 | 166,892 | 18% |
| Los Angeles | 21,323 | 398,972 | 5% |
| Merced | 875 | 13,292 | 7% |
| Riverside | 5,300 | 99,141 | 5% |
| San Diego | 17,776 | 263,825 | 7% |
| San Francisco | 13,385 | 151,027 | 9% |
| Santa Barbara | 27,418 | 91,704 | 30% |
| Santa Cruz | 9,514 | 70,762 | 13% |
| **Total** | 159,786 | 1,662,574 | 10% |

**Source:** American College and University Presidents' Climate Commitment Reporting System

**Notes:** 2012 GHGs total are gross scope 1, 2 and 3 GHGs combined.
Reporting years for Davis and LA are 2013; reporting year for Santa Cruz is 2011.

## UCLA Air Travel Emission Statistics Comparison over the years

| Topic | Item | 2007 | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |
| **Methodology** | AirTravel Emission Calculation | Calvin Method | Extrapolated | Extrapolated | Calvin Method | Extrapolated | Extrapolated | Calvin Method | Calvin Method |
| **Factors** | Flight broken by distance & fuel economy | yes | | | yes | | | yes | yes |
| | Aircraft model | yes | | | yes | | | yes | yes |
| | LTO cycles | yes | | | yes | | | yes | yes |
| | Load Factor | yes | | | yes | | | yes | yes |
| | RFI (effect of other GHGs) | no | | | no | | | no | no |
| | | | | | | | | | |
| | Faculty | 2,158 | 2,173 | 2,098 | 2,114 | 2,118 | 2,103 | 2,113 | 2,170 |
| **UCLA Population** | Staff | 18,464 | 19,122 | 19,957 | 18,881 | 18,478 | 17,946 | 19,561 | 19,985 |
| | Total Employee | 20,622 | 21,295 | 22,055 | 20,995 | 20,597 | 20,049 | 21,674 | 22,155 |
| | | LRDP | | AIMS | AIMS | | | | |
| **Survey Response Distribution:** | | | | | | | | | |
| | assistant prof | 56 | | | 124 | | | | |
| | associate prof | 59 | | | 77 | | | | |
| | distinguished prof | - | | - | 51 | | | | |
| | lecturer | 56 | | | 33 | | | | |
| | professor | 300 | | | 221 | | | | |
| | professor emeritus | - | | - | 5 | | | | |
| | Other (Instructors) | - | | - | 39 | | | | |
| | | | | | | | | | |
| | assistant prof | 0.12 | | | 0.27 | | | | |
| | associate prof | 0.13 | | | 0.17 | | | | |
| | lecturer | 0.12 | | | 0.07 | | | | |
| | professor | 0.64 | | | 0.49 | | | | |
| | | | | | | | | | |
| **Inventory** | UCLA Travel Services | 49,100,000 | | | 45,424,558 | 44,563,263 | 62,848,586 | 62,963,493 | 60,650,219 |
| **Air Travel Miles** | Faculty Travel via Survey | 17,700,000 | | | 24,850,811 | 24,900,650 | 24,719,631 | 34,816,668 | 35,755,878 |
| | Athletics department | 3,700,000 | | 207,670 | 8,561,359 | 8,561,359 | 8,561,359 | 9,076,840 | 9,942,492 |
| | # Faculty Short Survey responses | 471 | | | 455 | | | 201 | NA |
| | **Total UCLA related Air Travel Miles** | **70,567,237** | **74,430,056** | | **78,836,728** | **78,025,272** | **96,129,576** | **106,857,001** | **106,348,589** |
| | | | | | | | | | |
| **No. of flights** | UCLA Travel Services | 28,305 | | | 28,723 | | 41,423 | 39,907 | 38,767 |
| | Non-UCLA Travel Services | 3,166 | | | 7,950 | | 7,908 | 11,364 | 11,670 |
| | Athletics department | 3,997 | | | 9,724 | | 9,724 | 10,412 | 11,322 |
| | **Total flights** | **35,468** | **37,410** | **48,741** | **46,397** | **48,638** | **59,055** | **61,683** | **61,759** |
| | or | 36,584 | | | | | | | |
| **Aircraft type: B737 (domestic)** | 0.60 | 42,203,495 | | | 47,149,153 | 46,663,853 | 57,491,327 | 63,906,979 | 63,602,917 |
| **B777 (trans-atlantic)** | 0.21 | 14,635,949 | | | 16,351,077 | 16,182,778 | 19,937,688 | 22,162,602 | 22,057,156 |
| **B747 (trans-pacific)** | 0.19 | 13,727,793 | | | 15,336,498 | 15,178,641 | 18,700,561 | 20,787,420 | 20,688,516 |
| | | | | | | | | | |
| **# flights: B737** | 0.87 | 31,874 | | | 40,424 | | 51,452 | 53742 | 53808 |
| **B777** | 0.08 | 2,754 | | | 3,493 | | 4,446 | 4643 | 4649 |
| **B747** | 0.05 | 1,956 | | | 2,481 | | 3,157 | 3298 | 3302 |
| | | | | | | | | | |
| **Emissions (MT)** | UCLA Travel Services | **13,956** | **14,411** | **14,926** | **15,760** | **15,612** | **19,292** | **21,323** | **21,235** |
| | Non-UCLA Travel Services | | | | | | | | |
| | Athletics | | | | | | | | |
| | Total Campus | | | | | | | | |
| **Measure of Emissions** | $CO_2$ for all passengers | $CO_2$ | $CO_2$ | $CO_2$ | $CO_2$ | $CO_2$ | $CO_2$ | $CO_2$ | $CO_2$ |

## References


*Carbon Neutral Travel Policy*. (n.d.). Retrieved May 9, 2015, from Cornell University Sustainable Campus: http://www.sustainablecampus.cornell.edu/initiatives/carbon-neutral-travel-policy

*Daily Dashboard*. (2015, May 9). Retrieved from California Carbon Allowance Futures: http://calcarbondash.org/

*Ethics, Compliance and Audit Services*. (n.d.). Retrieved May 8, 2015, from University of California Office of the President: http://ucop.edu/ethics-compliance-audit-services/policy/policy-governance.html

*Greenhouse Gas Emissions Inventory*. (n.d.). Retrieved May 9, 2015, from Cornell University Sustainable Campus: http://www.sustainablecampus.cornell.edu/initiatives/greenhouse-gas-emissions-inventory

*Greenhouse Gas Emissions Report Fiscal Year 2013*. (2013). Retrieved May 5, 2015, from University of Arizona: http://rs.acupcc.org/site_media/uploads/ghg/212-2013-inventoryreports.pdf

Kwan, C. L. (2008, December 23). Quantifying the Carbon Footprint of UCLA related Air Travel. 3-58.

Le Quéré, C., Capstick, S., Corner, A., Cutting, D., Johnson, M., Minns, A., & Wood, R. (2015, March). Towards a culture of low-carbon research for the 21st Century. *Tyndall Centre for Climate Change Research Working Paper 161*.

*Optional Emissions from Commuting, Business Travel and Product Transport*. (2008, May). Retrieved from U.S. Environmental Protection Agency: http://www.epa.gov/climateleadership/documents/resources/commute_travel_product.pdf

*PCS Meeting MInutes*. (2014, January 29). Retrieved May 9, 2015, from Seattle University: https://www.seattleu.edu/uploadedFiles/Sustainability/Content/2014%20Jan%2029%20PCS%20meeting%20minutes.pdf

*The President's Committee on Sustainability*. (2011, October). Retrieved from Seattle University's Climate Action Tactical Plan FY 12 and 13.

*Toward a More Sustainable and Just World*. (2014). Retrieved May 9, 2015, from Seattle University: https://www.seattleu.edu/uploadedFiles/Sustainability/Content/SU%20Sustainability%20Progress%20Report%202014.pdf

*Treading Lightly*. (n.d.). Retrieved from University of Arizona Climate Action Plan: http://www.portal.environment.arizona.edu/sites/www.portal.environment.arizona.edu/files/portal/pdf/UA-CAP-Transportation.pdf

*Utah State University Carbon Offset Program*. (n.d.). Retrieved from Blue Goes Green: http://sustainability.usu.edu/htm/about/carbon-offset